

DATA MINING PROJECT
Master in Data Science and Advanced Analytics

NOVA Information Management School
Universidade Nova de Lisboa

Multi-Perspective Clustering Analysis for Loyalty Program Optimization at Amazing International Airlines Inc.

Group 45

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1. EXECUTIVE SUMMARY

This project delivers a data-driven customer segmentation to support Amazing International Airlines Inc. (AIAI) in improving the effectiveness of its loyalty program. By adopting two complementary perspectives (customer value and engagement, and travel behavior), three clear and actionable customer segments were identified, demonstrating that behavioral patterns are the primary drivers of customer value.

High-Value Core Customers generate a disproportionate share of value and should be prioritized through exclusive benefits, proactive service recovery, and highly targeted communications to protect and reinforce loyalty. The **Loyal and Engaged Majority** represents the largest segment, characterized by consistent activity and long tenure. Strategies for this group should focus on retention and incremental value creation through personalized offers, group-oriented rewards, and engagement-based incentives. **Low Engagement Customers** exhibit limited activity and contribution, and reactivation efforts for this segment should rely on simple, short-term incentives, price-led messaging, and low-cost engagement techniques.

Overall, this segmentation enables AIAI to allocate resources more efficiently and shift toward a targeted, behavior-driven loyalty strategy that maximizes long-term customer value.

2. INTRODUCTION

Building on the exploratory data analysis conducted in Phase 1, this stage focuses on translating previously identified patterns into actionable customer segments through clustering techniques. The initial analysis of Amazing International Airlines Inc.'s (AIAI) loyalty program data revealed a highly heterogeneous customer base, with strong imbalances in variables such as Income, Customer Lifetime Value (CLV), and flight activity. A small subset of customers was found to generate a disproportionate share of value, while the majority contributed more modestly.

From a behavioral perspective, significant variability in travel frequency and engagement was observed, with most customers flying infrequently and a smaller group of highly engaged frequent flyers accounting for a large share of flights and distance traveled. Seasonal patterns further highlighted the relevance of temporal and behavioral features for segmentation. These insights directly informed the data preparation and feature engineering steps, while previously identified data quality issues were addressed to ensure robust clustering results.

Within this context, the objective of the current phase is to apply and compare multiple clustering methodologies to identify distinct, interpretable customer segments based on loyalty, engagement, and travel behavior, supporting AIAI's transition toward more targeted, data-driven loyalty strategies.

3. METHODOLOGY

1. Data Preprocessing

1.1 Duplicates and Inconsistencies

Data preprocessing began with the removal of duplicate records from the flight_data dataset and the correction of inconsistencies identified in Phase 1. Zero values in NumFlights with positive DistanceKM were converted to missing, and inconsistent Promotion 2021 enrollments were reclassified as Standard.

1.1. Feature Engineering

Beyond the initial development of key performance indicators such as Activity Status, Tenure Years, Flights per Year, Frequency Category, and Typical Haul Type, further exploration of the data revealed opportunities to refine customer profiles. To address the excessive granularity of the Province/State variable, it was aggregated into five Canadian regions (Atlantic, Central Canada, Prairie Provinces, West Coast, and the North), reducing dimensionality while preserving meaningful regional travel patterns.

The analysis of NumFlightsWithCompanions revealed a strong right skew, with most observations equal to zero. To capture the key behavioral signal of whether customers travel with companions, this variable was transformed into a binary feature (HasCompanions). In parallel, for clustering purposes, the Companions_Ratio was introduced to retain information on the intensity of companion travel while mitigating statistical distortion.

After merging the datasets, CLV_per_Year was engineered by combining Customer Lifetime Value and Tenure Years. This variable measures annual customer value and enables fairer comparisons between long-tenured and newer members by accounting for loyalty program tenure. Overall, this Feature Engineering and EDA phase was critical for uncovering behavioral patterns not immediately visible in the raw data and for ensuring that the clustering process reflects meaningful customer dynamics rather than noise.

Our exploration of these new features provided the following final insights:

- Customers have an average tenure of nearly three years, with more than half exceeding two years in the loyalty program.
- Retention is strong, with only around 5% of customers having canceled their subscription.
- Despite 4,490 customers averaging zero flights per year, the customer base is primarily composed of frequent travelers rather than occasional users
- Medium-haul routes are preferred by most customers, while variability in travel distances enables meaningful segmentation.
- The customer base displays diverse travel behaviors, with clear distinctions between solo travelers and leisure or family-oriented travelers.
- Most customers generate low annual value, while a small subset shows exceptionally high CLV_per_Year.

1.2. Addressing Outliers

A rigorous outlier analysis was conducted to prevent extreme values from distorting the clustering results. A hybrid approach was adopted, combining univariate outlier detection using the Interquartile Range (IQR) method with expert business judgment, followed by a multivariate outlier analysis.

1. Univariate Outlier Handling

Our standard approach was IQR method ($1.5 * IQR$) to our variables for both datasets. This successfully identified and removed statistical noise without sacrificing significant sample size.

Some exceptions should be noted such as Customer Lifetime Value (CLV), NumFlightsWithCompanions, Companions_Ratio, PointsRedeemed, and DollarCostPointsRedeemed, where the standard IQR approach would have resulted in losses well above the 5% rule of thumb. Recognizing both the magnitude of this loss and the importance of these variables for capturing meaningful loyalty behavior, we opted for a manual thresholding cap based on visual inspection of the distributions, removing only the most extreme values.

2. Multivariate Outlier Handling

This step was intentionally performed after data merging, aggregation, and scaling to reduce computational complexity. Using DBSCAN on the aggregated dataset, we identified multivariate outliers, customers whose profiles are anomalous across the full feature space rather than within a single variable.

Based on the k-distance plot ($k = 16$), an elbow was observed at $\epsilon \approx 1.0$ ([Annex 1](#)), which was selected as the threshold. The model identified 16,090 core customers forming dense and stable clusters, while 439 customers were classified as noise. These outliers were retained for post-clustering classification to prevent distortion of the clustering results.

1.3 Missing Values

A small number of missing values were identified in the *Income* and *Customer Lifetime Value* variables, including those created to address the initial inconsistency. These observations were handled during the univariate outlier treatment, and as a result, no further imputation was required.

1.4 Feature Selection

The feature selection phase focused on retaining variables that add meaningful value to customer segmentation while reducing redundancy, noise, and model complexity, guided by variability, business relevance, and correlation structure. The first step consisted of identifying constant or quasi-constant variables, defined as variables with less than 1% variability:

- Country contained only the value Canada for all observations. As a result, it was removed from further analysis.

Several variables were excluded due to their limited relevance for clustering and cluster profiling:

- First Name, Last Name, Customer Name: Removed as they provide no analytical value for segmentation
- Postal Code: Excluded due to its high granularity and limited interpretability for clustering.
- EnrollmentDateOpening: Removed since customer tenure is already captured by the engineered Tenure_Years variable, which is more relevant for segmentation.
- CancellationDate: Considered redundant because churn information is represented by Activity_Status, and it contains missing values for active customers.
- YearMonthDate: Excluded as its information is already captured by Year and Month.

To identify redundant variables, correlation analysis was performed on the numerical features, as illustrated in the correlation heatmaps ([Annex2](#)).

Given this redundancy, a correlation threshold of 0.75 was applied, and variables exceeding this limit were removed. As a result, DistanceKM was dropped in favor of Typical Haul Type, PointsAccumulated was excluded while DollarCostPointsRedeemed was retained for its stronger business relevance, and both HasCompanions and NumFlightsWithCompanions were removed as their information is already captured by the Companions_Ratio variable.

1.5 Dataset Aggregation and Merging

Before clustering, all datasets were aligned to a consistent customer-level granularity. As the flights dataset contains multiple observations per customer, it was aggregated so that each row represents a unique customer identified by Loyalty#. Cumulative variables were summed to preserve engagement intensity, while categorical behaviors were aggregated using the mode to reflect typical behavior. Engagement duration was captured through active years and months. The aggregation results are presented in ([Annex 3](#)).

1.6 Scaling

Before applying clustering algorithms, given the presence of significant outliers in variables such as CLV and flight-related metrics, which ended up being intentionally retained due to their strategic value, the RobustScaler was selected for feature scaling. By centering the data based on the median and scaling according to the Interquartile Range (IQR), we ensured that the resulting feature scales are robust to extreme observations, preventing them from disproportionately dominating the distance calculations in our multivariate analysis.

2. Clustering

For the clustering analysis, two complementary segmentation perspectives were defined. The **Customer Value & Loyalty Engagement** (CVLE) perspective focuses on economic contribution and relationship maturity, incorporating income, lifetime value, loyalty point activity, and tenure-based metrics, including the normalized CLV per year. In parallel, the **Travel Behavior** (TB) perspective captures how customers interact with the airline, emphasizing flight frequency, companion travel intensity, and engagement duration.

2.1 Hierarchical Clustering

We applied Hierarchical Clustering using Ward linkage for the **CVLE perspective**, as it showed the highest and most consistent R^2 values across the tested solutions ($k = 1$ to $k = 10$) in the variance-explained plot ([Annex 4](#))

Based on the dendrogram ([Annex 5](#)), the tree was cut at $k = 4$, while $k = 5$ was also evaluated for comparison. For this and all subsequent comparisons, cluster validity was assessed using three metrics: the Silhouette Score, the Calinski–Harabasz Index, and the Davies–Bouldin Index.

In addition to the Silhouette Score, clustering quality was evaluated using the Calinski–Harabasz (CH) and Davies–Bouldin (DB) indices. The CH Index measures the ratio of between- to within-cluster dispersion, with higher values indicating more compact and better-separated clusters. The DB Index measures the average similarity between each cluster and its most similar cluster, where lower values indicate stronger separation and better overall clustering performance.

The metric results (see [Annex 6](#)) indicated that $k=4$ provides slightly stronger separation and compactness than $k=5$.

For the **Travel Behaviour** solution, $k = 2$ and $k = 3$ were evaluated and compared using the same validation metrics (see [Annex 7](#) and [Annex 8](#)). The 2-cluster solution clearly outperformed, indicating better cohesion and separation. As a result, $k = 2$ was selected as the most stable and interpretable solution for this perspective within the HC.

2.2 K-Means

Then, **K-Means** was applied as a complementary approach to evaluate the robustness of the segmentation results. For the **CLVE perspective**, cluster solutions ranging from $k = 2$ to $k = 10$ were assessed using the Silhouette Score. Both the Inertia Plot and the Average Silhouette Plot indicated $k=2$ as the most appropriate solution (see [Annex 9](#) and [Annex 10](#) respectively). This $k=2$ solution was therefore selected, as it provides a balance between clustering quality, interpretability, and managerial usefulness. For the **TB perspective**, an identical validation approach was adopted, relying on the Inertia Plot ([Annex 11](#)) and the Silhouette Score ([Annex 12](#)). Once again, the $k = 2$ solution outperformed higher-cluster alternatives and was retained as the most stable and interpretable segmentation.

2.3 SOM

Then, a **Self-Organizing Map** was implemented as a two-stage clustering method, as it has advantages in computational efficiency and pattern preservation. SOMs project customers into a 2D grid while preserving neighborhood structure, enabling clearer visual assessment of segment boundaries and variable-driven patterns.

For the **CLVE perspective**, the SOM was tuned to balance interpretability and topological preservation. A 9×9 hexagonal grid with a Gaussian neighborhood ($\sigma = 1.5$) and a learning rate of 0.05 was selected. Batch training was conducted for 80,000 iterations (five times the number of

observations) to ensure stability and convergence, with the learning rate reduced from 0.5 to prevent weight instability. This configuration achieved a Quantization Error of 0.6484 (a 10.7% improvement) and reduced the Topographic Error to 0.1582 (an 83.4% improvement), confirming effective representation of the data and preservation of the dataset's topological structure.

To visually validate our interpretations, three complementary visual outputs were analyzed. The component planes ([Annex 13](#)) are well organized, showing smooth gradients and consistent patterns across correlated variables such as Customer Lifetime Value and CLV_per_Year, with a clearly defined region representing a small high-value customer subgroup. PointsAccumulated is high across most of the map, whereas DollarCostPointsRedeemed is concentrated in a specific area, indicating that point accumulation is widespread but redemption is limited to a subset of customers. Tenure_Years spans both high- and low-value regions, suggesting that tenure alone does not define customer value, while Income displays a diffuse pattern, reinforcing that value is driven primarily by behavioral rather than income-related factors.

The U-matrix ([Annex 14](#)) reveals a well-defined structure, with a large homogeneous area (dark blue) representing groups of customers that are very similar to one another. Surrounding this region, lighter cells act as transition zones, indicating customers that lie between profiles. At the extremes of the map, higher-distance neighbourhoods highlight neurons that are highly dissimilar, suggesting clear boundaries between distinct customer segments.

Finally, the SOM hits map ([Annex 15](#)) shows that most observations are concentrated in a small number of neurons, while the neurons associated with high CLV and CLV per Year, as identified in component planes receive comparatively fewer hits. These patterns confirm that high-value customers represent a minority within the overall population.

For the TB perspective, a 7×7 hexagonal grid was selected to provide adequate resolution for the lower-dimensional feature space, using a Gaussian neighborhood function with $\sigma = 2.0$ and a learning rate of 0.5. After training, both Quantization Error (QE) and Topographic Error (TE) decreased, corresponding to improvements of 0.4% and 96.8%, respectively. Model tuning focused primarily on grid size, with multiple configurations tested until the lowest topological error was achieved.

Visual inspection of the component planes ([Annex 16](#)) also shows good organization and smooth patterns. We can highlight a positive correlation between Companions_ratio and Flights_Per_Year, indicating that customers who travel more frequently also tend to travel with companions. In contrast, Tenure_Years follows a smoother and independent gradient, suggesting that tenure doesn't directly translate to higher travel activity. The U-matrix ([Annex 17](#)) highlights a big distance boundary on one side of the map, suggesting a strong separation between two main travel behaviour profiles. This structure is also reinforced by the SOM hits map ([Annex 18](#)), which shows that most customers are concentrated in a limited number of neurons, while the more distant region contains fewer but behaviourally distinct customers.

After reducing the dimensionality of the CLVE and TB datasets, both K-Means and Hierarchical Clustering were applied to the SOM units. This procedure follows the same logic

described in Sections 2.2 and 2.1, respectively, with the key difference that it clusters SOM neurons with similar profiles rather than individual observations.

2.3.1 SOM + K-Means

For the **CLVE perspective**, the inertia plot ([Annex 19](#)) and the silhouette score ([Annex 20](#)) indicated $k = 2$ as the optimal solution. Although a two-cluster solution may appear overly simple, higher values of k led to a noticeable decrease in the silhouette score, indicating that additional clusters increased fragmentation without meaningful improvements in clustering quality.

For the **TB perspective**, the Inertia Plot ([Annex 21](#)) shows a clear elbow on $k = 2$, which is also the solution with the highest silhouette score ([Annex 22](#)).

2.3.2 SOM + Hierarchical Clustering

We also tested Hierarchical Clustering with Ward linkage on the SOM prototypes to identify the best-performing model.

For the **CLVE perspective**, the dendrogram ([Annex 23](#)) suggested two potential solutions at $k = 3$ and $k = 4$. Both configurations were therefore estimated and compared using the previously defined validation metrics. While both solutions produced coherent clusters, the four-cluster solution achieved better overall performance, with the highest silhouette score and the lowest Davies–Bouldin Index, and was consequently selected. ([Annex 24](#))

In the **TB perspective**, the dendrogram ([Annex 25](#)) lead us to explore $k=2$ and $k=3$, with $k=2$ outperforming the other solution. ([Annex 26](#))

2.3.3 K-Means VS Hierarchical Clustering

Within the SOM framework, **K-Means** and **Hierarchical Clustering** were first compared across the candidate solutions for each perspective.

For the **CVLE perspective**, SOM + K-means was preferred due to its more balanced performance across metrics all metrics ([Annex 27](#)). For the **TB perspective**, SOM + K-Means achieved best results. ([Annex 28](#))

2.4 Final Solution for Each Perspective

Finally, the 3 approaches were compared to obtain the final model: K-means, Hierarchical Clustering, SOM + K-means, and SOM + Hierarchical Clustering. For **CLVE**, the results show that **SOM + K-Means** for a $k=3$ achieved the strongest overall clustering performance across all validation metrics, and was therefore selected as the final solution ([Annex 29](#)). For **TB**, all three approaches produced similar silhouette scores, but K-Means ($k = 2$) achieved the highest silhouette score and Calinski-Harabasz (see [Annex 30](#)), indicating slightly better cluster compactness.

Despite the quantitative comparisons above, relying solely on internal validation metrics can be limiting, as they capture only numerical aspects of cluster structure. In line with the No Free Lunch

Theorem, no single clustering method performs best in all contexts. Therefore, model selection was guided not only by validation metrics but also by interpretability, stability, and alignment with AIAI's business objectives.

The final solutions were combined by computing centroid profiles for each cluster intersection. Hierarchical clustering (Ward linkage) was then applied to these centroids, resulting in a final solution with three merged clusters, explaining 43.9% of the variance.

3. Cluster Visualization

For cluster visualization, PCA, t-SNE, and UMAP were applied to project the data into two dimensions. Together, these methods provide qualitative support for the final three-cluster solution ([Annex 31](#)). While PCA shows some overlap between clusters, both t-SNE and UMAP display clear separation, indicating that the segments are not linearly separable. The t-SNE projection reveals well-defined cluster regions, confirming coherent customer similarity patterns, while UMAP further reinforces clear cluster differentiation. Both non-linear methods reveal some degree of fragmentation, suggesting that Cluster 2 (green) is more heterogeneous, potentially containing closely related but not fully uniform profiles, whereas Cluster 1 appears more compact and consistent. Overall, these visualizations confirm that the identified clusters represent distinct customer profiles, with non-linear techniques capturing the underlying structure more effectively than linear projections.

In the case of PCA, a trade-off between simplicity and information loss was observed. The first two principal components captured 66.6% of the total variance, implying an information loss of 33.4%. The first principal component was primarily driven by Customer Lifetime Value (0.63) and CLV_per_Year (0.76), and was therefore interpreted as Customer Value. The second principal component reflected Engagement/Customer Activity, as it was mainly influenced by PointsAccumulated, Companions_Ratio, Flights_Per_Year, DollarCostPointsRedeemed, and Tenure_Years ([Annex 32](#)).

To tune t-SNE, a perplexity value of 30 was selected. After testing multiple values, higher perplexity levels produced minimal changes, and additional runs with different random seeds confirmed the stability of the projection. For UMAP, the configuration used 15 neighbors and a min_dist of 0.1, which was sufficient to represent the distances between clusters.

4. Feature Importance and Classification of Multivariate Outliers

Feature importance was assessed using two complementary approaches: an R^2 -based measure ([Annex 33](#)) and Gini importance derived from a Decision Tree. While the R^2 metric highlights variables that differ most across clusters, Gini importance identifies features that most effectively split the data through decision rules. Both approaches consistently indicate that Flights_Per_Year and CLV_per_Year are the most influential variables, whereas Income and several other engagement-related features contribute limited additional information according to Gini importance.

Finally, the Decision Tree was used to assign the outliers previously identified by DBSCAN to the existing three clusters by predicting their most likely segment. This step allowed the observations

set aside during preprocessing to be integrated into the final segmentation, ensuring that potentially meaningful customer profiles were retained without distorting the clustering algorithms.

4. RESULTS & VALIDATION

The resulting clusters were then analyzed to assess their coherence and interpretability from a business perspective. Based on the parallel coordinates plot, the following cluster labels were defined ([Annex 34](#)):

Cluster 0 – High-Value Core

Cluster 0 represents the smallest cluster but clearly the most valuable. It stands out with very high Customer Lifetime Value and CLV per Year, while showing below-median Points Accumulated and Flights per Year, suggesting fewer but highly profitable customers. Tenure is around the average, indicating that high value is driven more by spending efficiency than by longevity or frequency.

Cluster 1 – Loyal and Engaged Majority

This cluster represents the largest share of customers and it is characterised by its above-median Tenure Years, moderate CLV, and higher engagement indicators, including Points Accumulated, Flights per Year, and Companions Ratio. This profile reflects loyal, consistently active customers who engage frequently but do not generate extreme value individually.

Cluster 2 – Low-Engagement / Low-Value Segment

Customers in this cluster show below-median performance across most metrics, including CLV, CLV per Year, Flights per Year, and Companions Ratio. Despite some accumulated points, overall engagement and value remains low. This segment represents more transactional or disengaged customers with limited tenure in general.

Overall, most non-metric variables represent very similar distributions across the 3 final segments, indicating that the segmentation is primarily driven by behavioral and value-related features, rather than socio-demographic characteristics.

Education level, marital status, and enrollment type exhibit nearly identical distributions across all segments ([Annex 35](#), [Annex 36](#), [Annex 37](#), respectively), indicating limited discriminative power for segmentation. Similarly, loyalty status and activity status show only minor variation between clusters, with all segments containing comparable proportions of Aurora, Nova, and Star members ([Annex 38](#)) and being predominantly composed of active customers ([Annex 39](#)).

In contrast, travel-related categorical variables provide clearer differentiation between segments. Flight frequency categories reveal meaningful separation, with frequent flyers strongly characterizing Cluster 1. Cluster 0 displays a more balanced distribution, encompassing a mix of frequent, moderate, and low-activity travelers, while Cluster 2 is clearly dominated by inactive or low-activity customers ([Annex 40](#)).

Finally, across all segments, medium-haul travel is the most prevalent. However, Cluster 2 stands out due to a noticeably higher proportion of no-haul and short-haul trips, whereas Cluster 1 is almost exclusively composed of medium-haul travelers ([Annex 41](#)).

5. STRATEGIC RECOMMENDATIONS

Our strategic recommendations are grounded in research on airline and travel-industry loyalty trends, drawing on insights from leading sources including McKinsey & Company, BCG, OAG, and Forbes. These insights inform targeted improvements to AIAI's loyalty program that align with proven best practices while addressing the specific needs of AIAI's customer segments.

Cluster 0 – High-Value Core

Loyalty Program Optimization

- Introduce **exclusive, experience-focused rewards** like VIP airport services, premium seat options, or curated travel packages, leveraging McKinsey's insight: *"benefits that go beyond points and rewards...are far stickier and can generate new value"* (Alderman et al., 2024).
- Implement **loyalty-only fare visibility or quiet pricing**, allowing elite members to see special fares unavailable to the public, following McKinsey's example of under-the-hood special pricing (Alderman et al., 2024).
- Offer **digital or onboard content perks** (e.g., free Wi-Fi, premium entertainment bundles) to enhance the travel experience, mirroring Delta SkyMiles' strategy of using experiential benefits to boost engagement (Chapple et al., 2023).

Service Customization

- Deploy **tier-based proactive service recovery**, so high-value members automatically receive compensation or apology offers aligned with their status, reflecting McKinsey's finding that service challenges can increase satisfaction if handled appropriately (Chapple et al., 2023).
- Use **predictive rebooking and disruption alerts**, giving VIP members automated support and personalized notifications to reinforce their status.

Marketing & Activation

- Limit communications to **highly targeted, low-frequency messages** but increase **exclusive, highly relevant opportunities** such as first access to new routes or flight classes, premium cabins, or exclusive offers to reinforce perceived status and exclusivity.

Cluster 1 – Loyal and Engaged Majority

Loyalty Program Optimization

- Introduce **travel streak challenges** that reward consistent flying across months or seasons, inspired by engagement-based gamification in airline loyalty programs (Bloom, 2025).
- Offer **group milestone rewards** for customers who frequently travel with companions, reflecting the cluster's higher companion ratios.

- Implement **points pooling and shared reward goals**, allowing families or friends to combine points toward a single redemption, reducing redemption friction and encouraging collective engagement (Bloom, 2025).

Service & Experience

- Offer **group-focused travel bundles** (seating, baggage, boarding) tailored to repeat and companion travelers.
- Use analytics to **anticipate repeat routes and seasonal travel**, enabling proactive and relevant service offers (McKinsey & Company, 2023).

Marketing & Activation

- Deploy **AI-driven offers based on repeat travel behavior**, such as frequently flown routes or recurring travel periods, following personalization practices highlighted by McKinsey & Company (Chapple et al., 2023).
- Provide **early access to promotions for long-tenured customers**, recognising sustained engagement rather than high spend alone.
- Promote **travel-relevant partner benefits** (hotels, rideshare, mobility) that support frequent trips, consistent with BCG's (2024) findings on partnership-driven loyalty.

Cluster 2 - Low Engagement

Loyalty Program Optimization

- Introduce **first-win rewards**, offering a small guaranteed benefit after the next flight to demonstrate immediate program value.
- Use **short-term earning boosts**, such as limited-time bonus points, to encourage a second or third booking.
- Apply **use-it-soon rewards** with short expiry periods to prompt action rather than long-term saving.

Service & Experience

- Offer **trial convenience perks**, such as one-time free seat selection or priority boarding, to showcase added value.
- Focus on **low-friction digital service**, including app-based check-in, boarding passes, and self-service support.
- Maintain **basic service reliability**, prioritising consistency over personalised or premium experiences.

Marketing & Activation

- Deploy **flash offers** triggered by browsing or abandoned searches to stimulate short-term demand.
- Use price-led messaging that highlights affordability, simplicity, and ease of booking.

6. CONCLUSION

This project demonstrates how a multi-perspective clustering approach can support AIAI in developing more targeted and value-driven loyalty strategies. By combining customer value, engagement, and travel behavior features, three distinct and interpretable segments were identified, showing that behavioral patterns, rather than socio-demographic characteristics, are the main drivers of customer value. The use of multiple clustering techniques (K-Means, Hierarchical Clustering, SOM-based approaches) and validation metrics (Silhouette Score, Calinski–Harabasz Index, Davies–Bouldin Index, and R^2) enhances the robustness and relevance of the final segmentation.

Regarding methodological limitations, this analysis relies exclusively on customers available in the database, which may introduce bias related to new, long-tenured, or inactive customers, as these groups can exhibit incomplete behavioral patterns that affect segment stability and interpretation. Additionally, aggregating the data at the customer level reduced the ability to capture seasonal variations in travel behavior. Finally, the results are sensitive to modeling decisions such as feature selection and parameter configuration, meaning that alternative choices could lead to different segmentation outcomes.

Future research could extend this work by integrating additional data sources related to customer satisfaction, such as survey-based feedback. Combining value-based features with customer feedback would improve the explanatory power of the segmentation and support more personalized strategies. Additionally, the use of A/B testing could be applied to validate and refine the segmentation through real business implementation. This would allow external validation of the cluster decisions and enable the quantification of the impact of the proposed strategies on customer engagement and profitability, supporting continuous improvement of the segmentation model.

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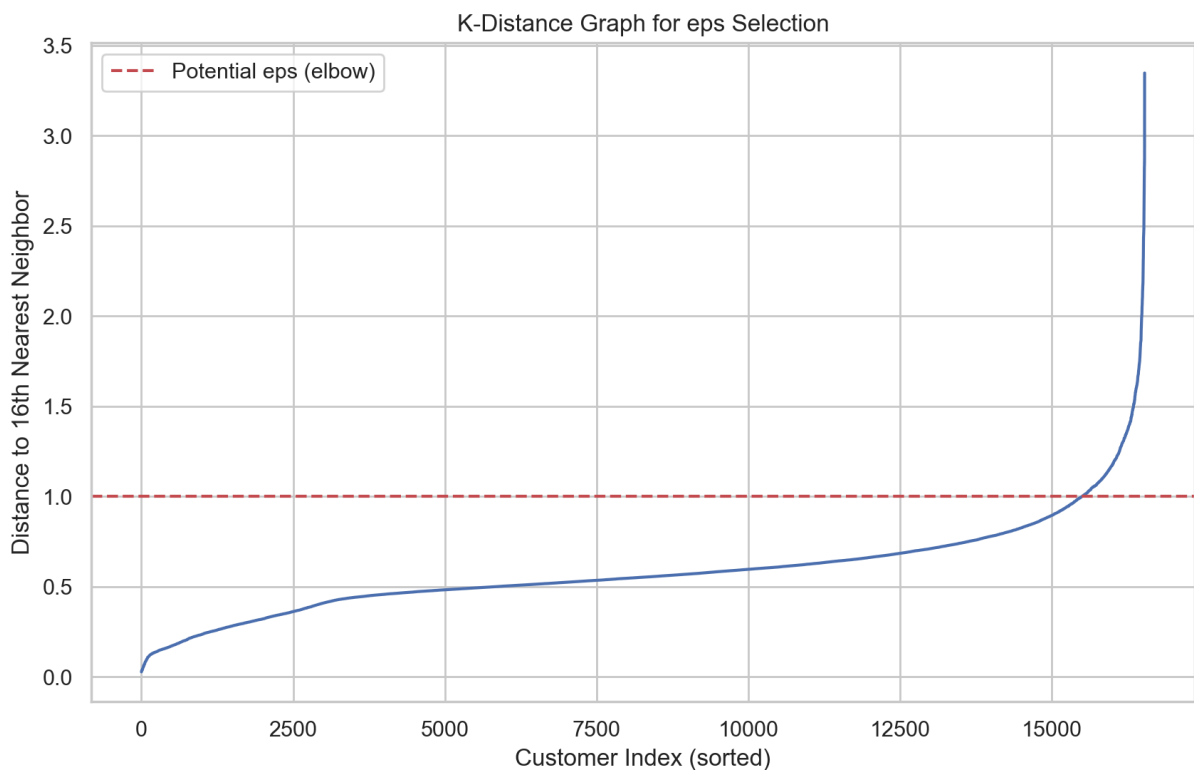
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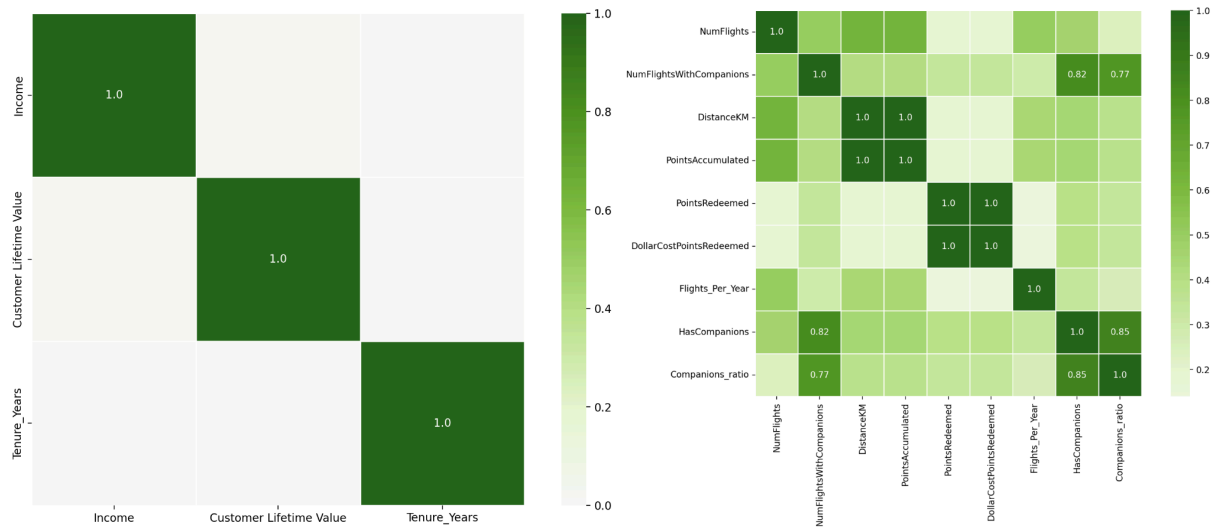
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8. ANNEXES

Annex1



Annex 2

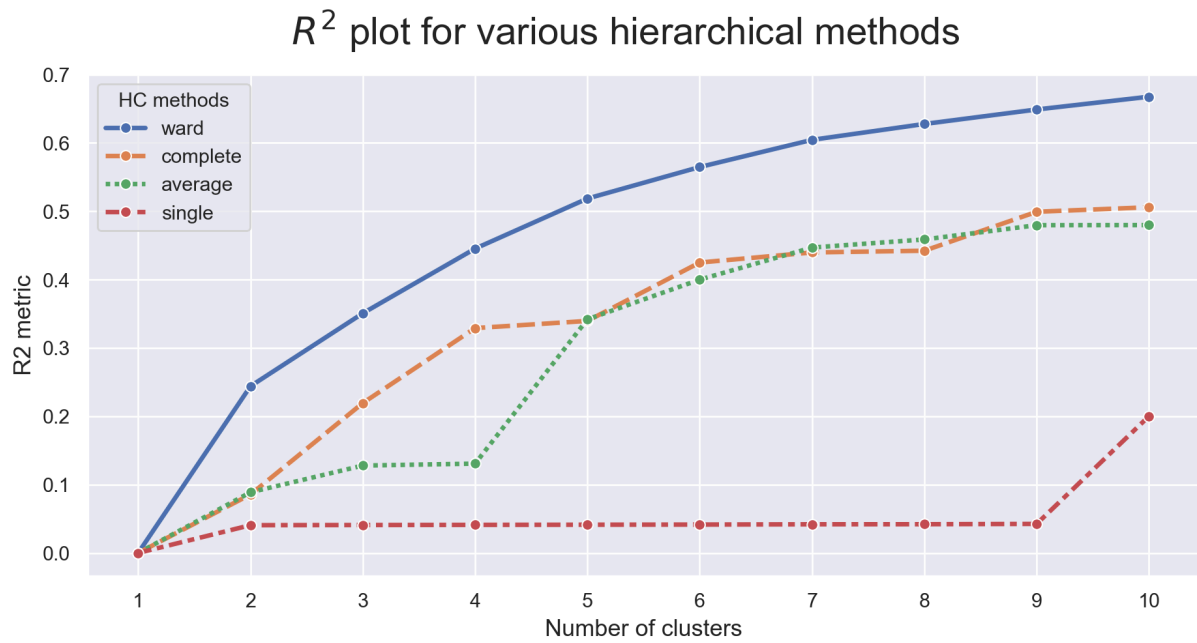


Annex3

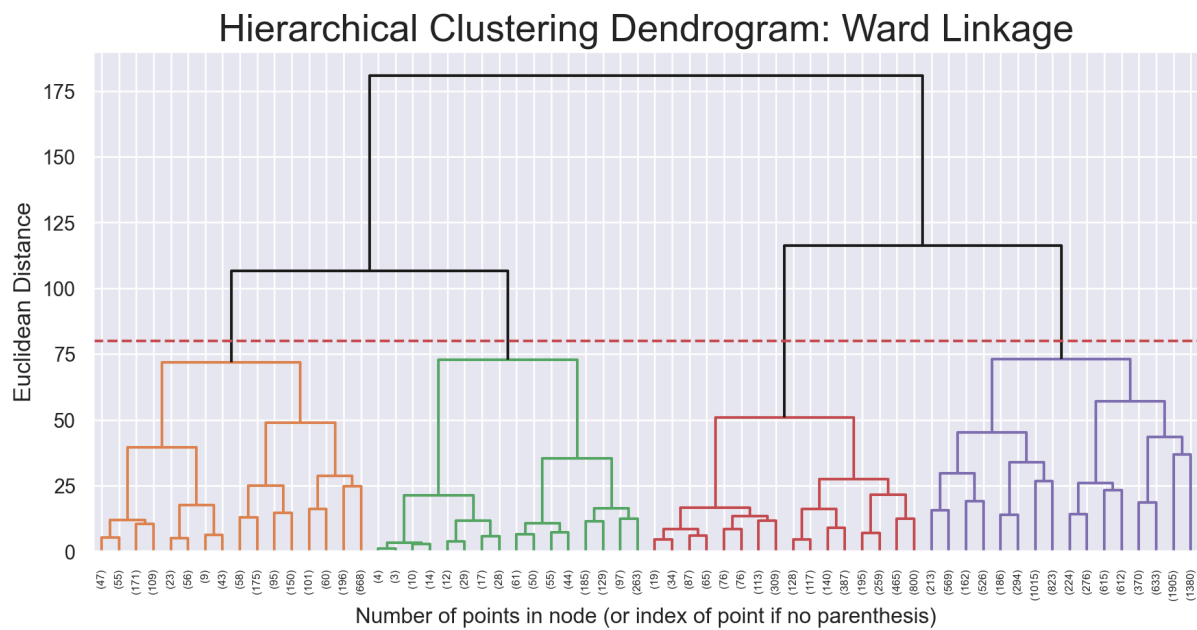
Results with aggregation:

- NumFlights — Total flights per customer
- CompanionsRatio - overall tendency to travel with companions across recorded trips
- PointsAccumulated — Total points earned
- DollarCostPointsRedeemed — Total monetary value redeemed
- FlightsPerYear- Mean of yearly flight totals per customer
- Frequency_Category - Each customer's most representative travel frequency segment
- Typical Haul Type - Customer's most common route distance behavior
- Years_Traveled - Number of unique years in which the customer had at least one flight
- Most_Active_Month - Modal month of activity

Annex 4



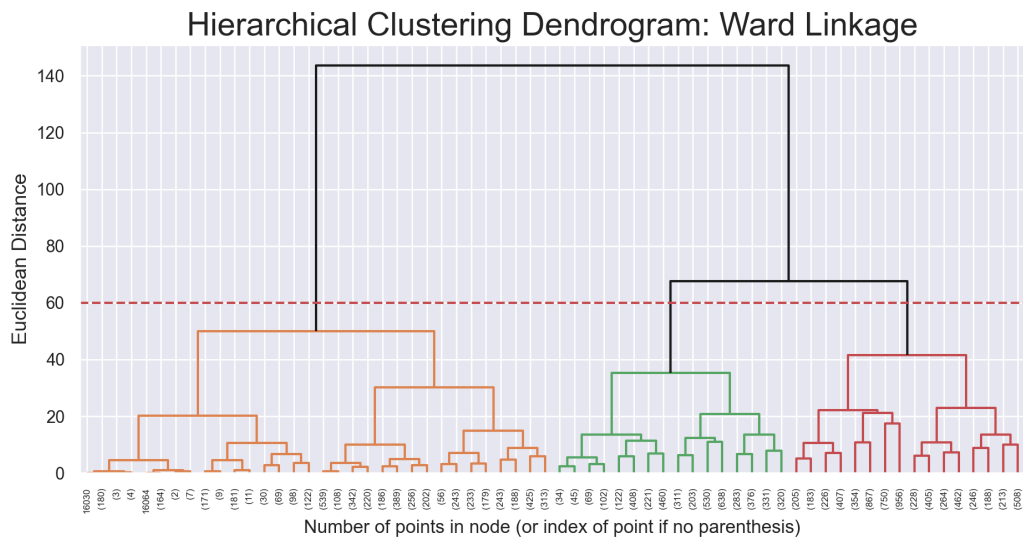
Annex 5



Annex 6

	Number of Clusters (k)	Silhouette Score (↑ better)	Calinski-Harabasz Index (↑ better)	Davies-Bouldin Index (↓ better)	Preferred
0	4	0.250	4824.31	1.27	Selected
1	5	0.188	4314.18	1.60	Not Selected

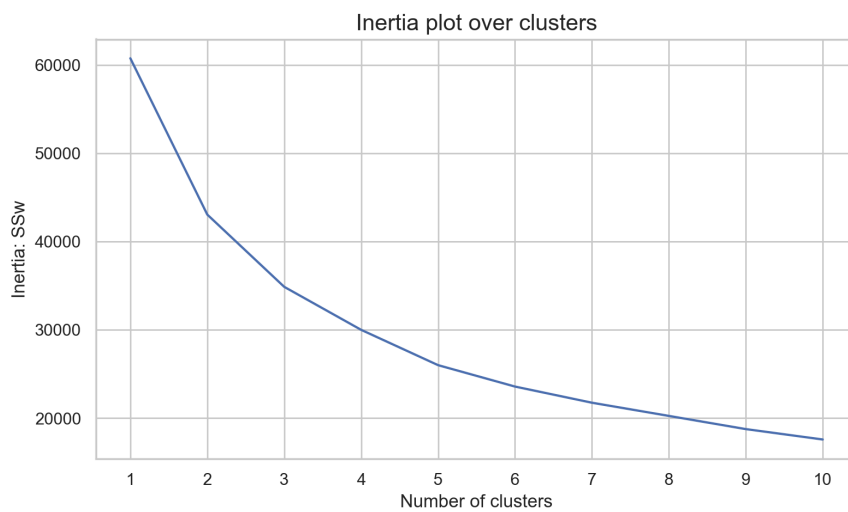
Annex 7



Annex 8

	Number of Clusters (k)	Silhouette Score (↑ better)	Calinski-Harabasz Index (↑ better)	Davies-Bouldin Index (↓ better)	Preferred
0	2	0.479374	17891.381746	0.775863	Selected
1	3	0.353559	14503.215812	1.102099	Not Selected

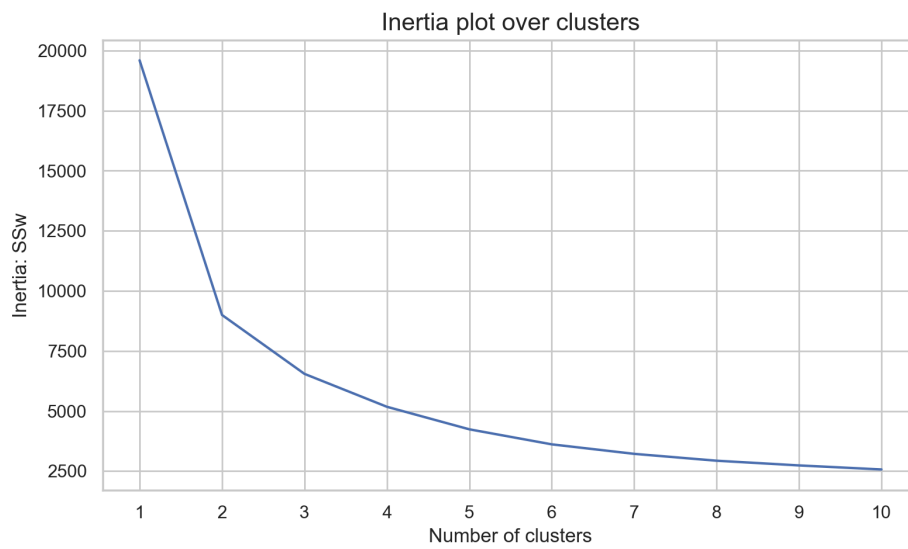
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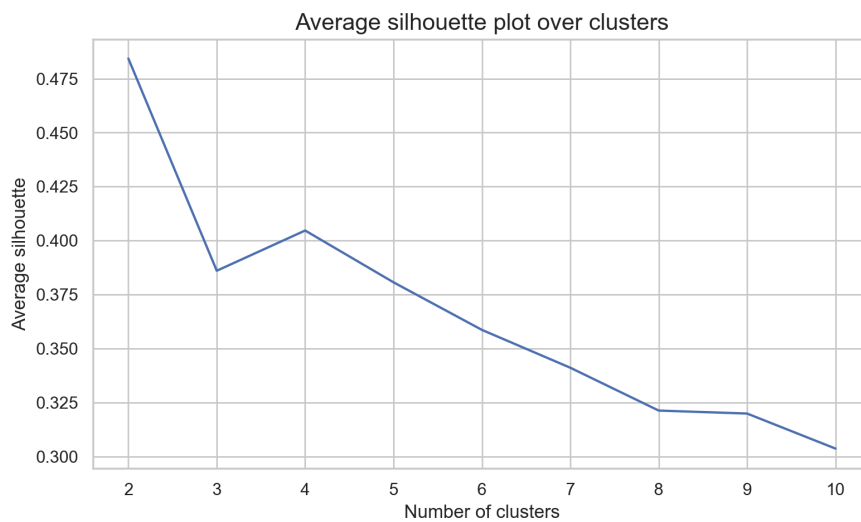
Annex 10



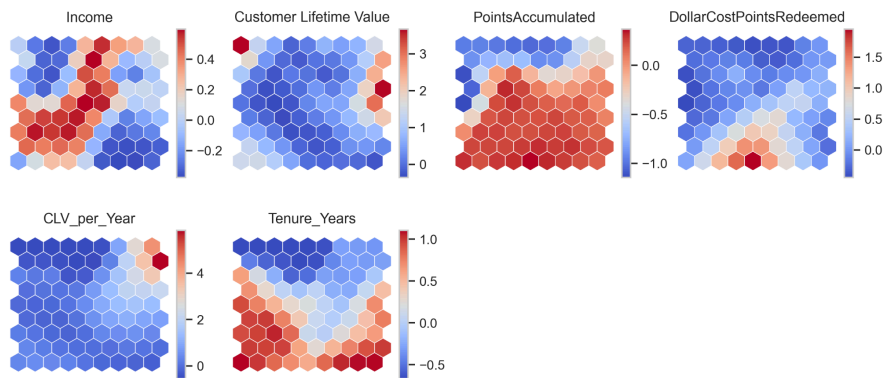
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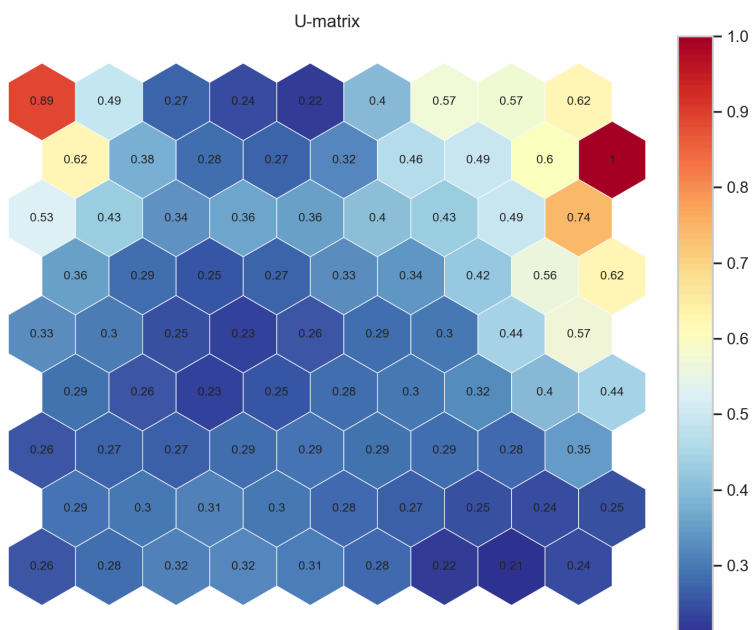
Annex 12



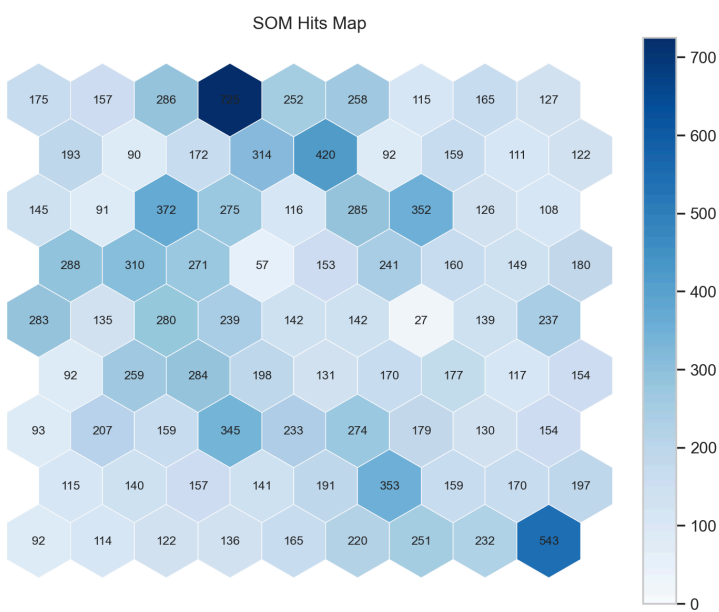
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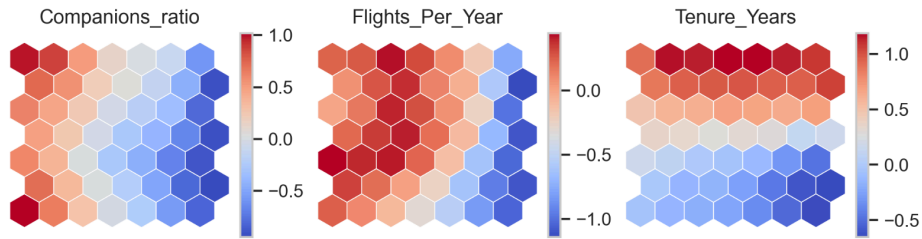
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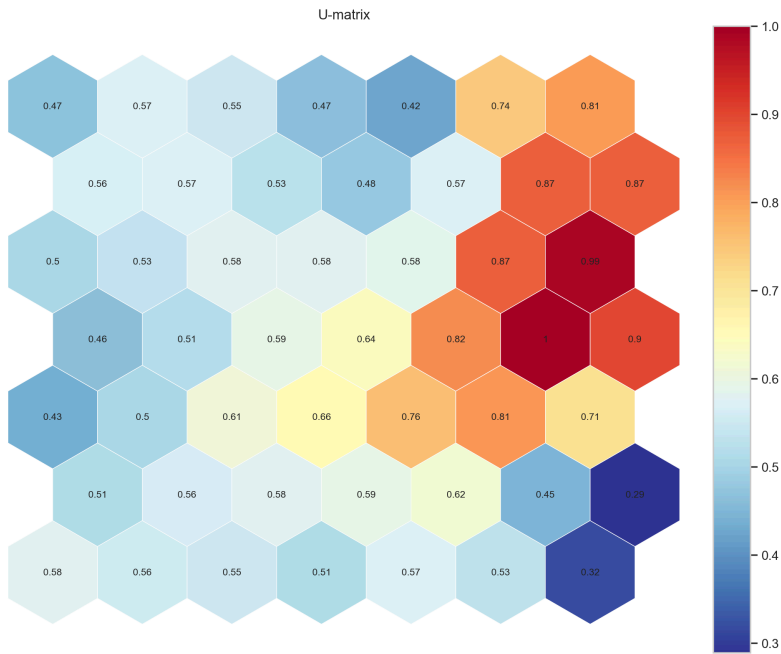
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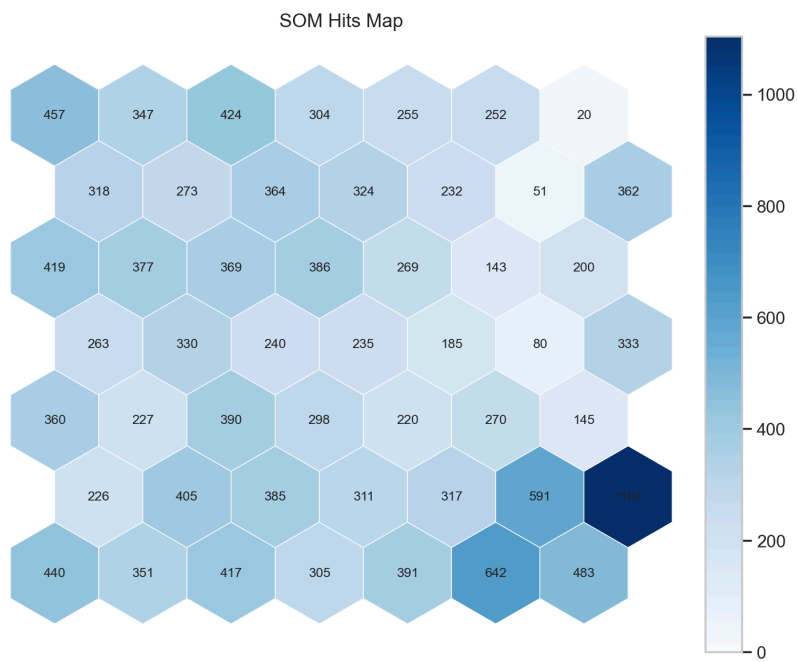
Annex 16



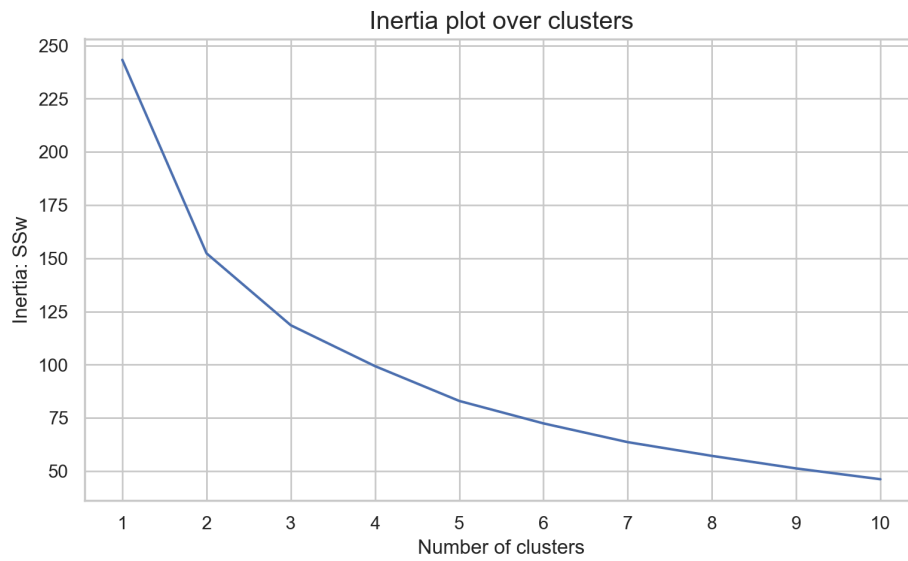
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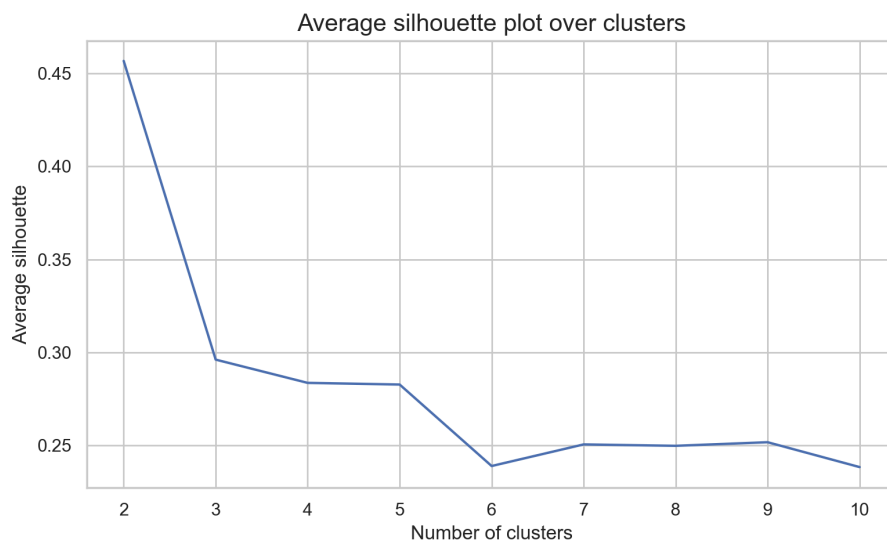
Annex 18



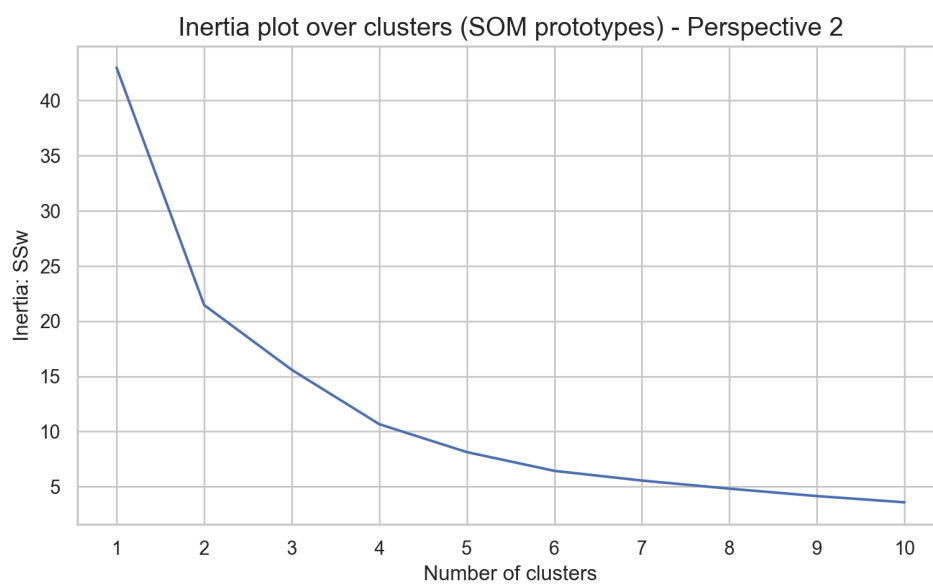
Annex 19



Annex 20



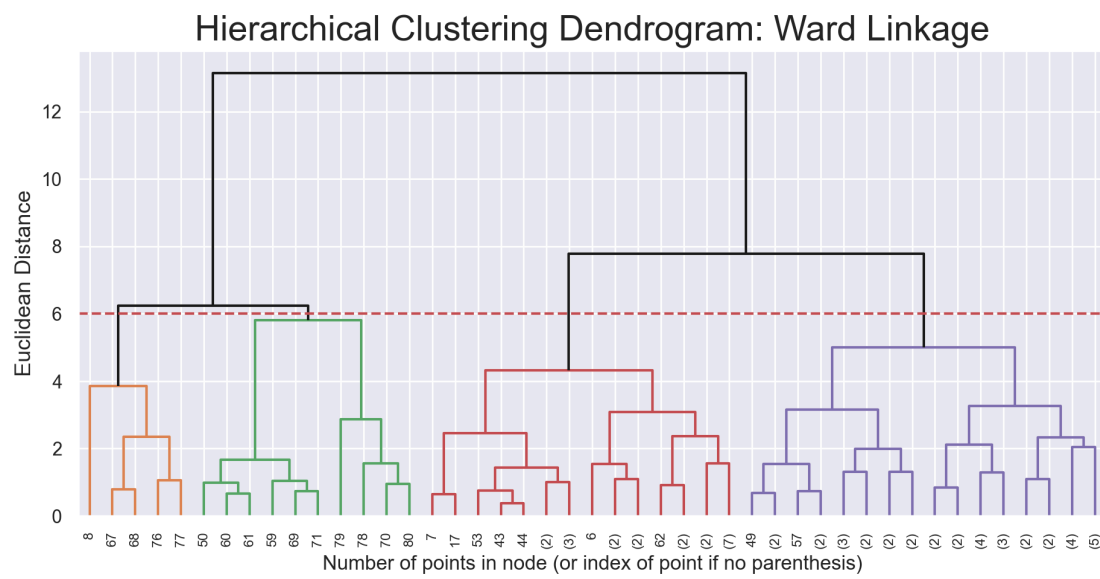
Annex 21



Annex 22



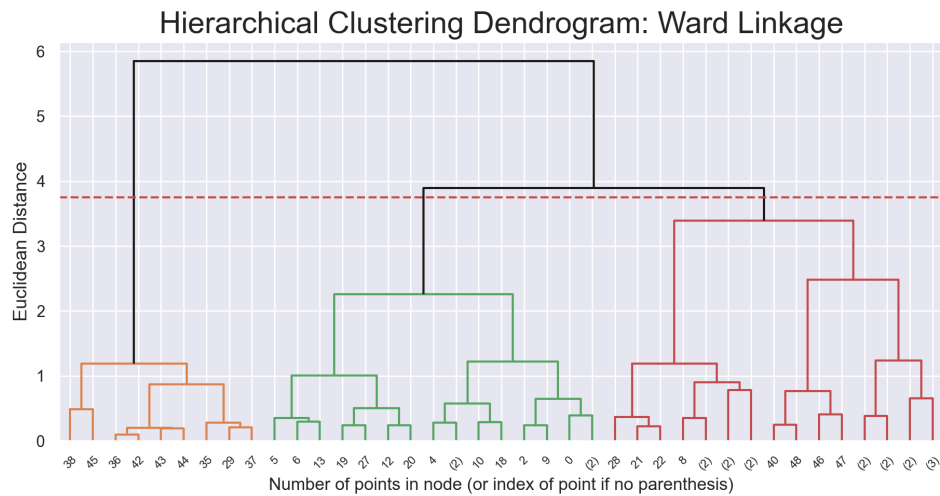
Annex 23



Annex 24

Number of Clusters (k)	Silhouette Score (↑ better)	Calinski-Harabasz Index (↑ better)	Davies-Bouldin Index (↓ better)	Preferred	
0	4	0.244	32.658	1.295	Selected
1	3	0.240	35.997	1.316	Not Selected

Annex 25



Annex 26

	Number of Clusters (k)	Silhouette Score (↑ better)	Calinski-Harabasz Index (↑ better)	Davies-Bouldin Index (↓ better)	Preferred
0	2	0.405	31.138	0.713	Selected
1	3	0.270	31.068	1.180	Not Selected

Annex 27

	Algorithm	Number of Clusters (k)	Silhouette Score (↑ better)	Calinski-Harabasz Index (↑ better)	Davies-Bouldin Index (↓ better)	Preferred
0	K-means	3	0.457	47.154	0.929	Selected
1	Hierarchical Clustering	4	0.244	32.658	1.295	Not Selected

Annex 28

	Algorithm	Number of Clusters (k)	Silhouette Score (↑ better)	Calinski-Harabasz Index (↑ better)	Davies-Bouldin Index (↓ better)	Preferred
0	K-means	2	0.425	47.103	0.925	Selected
1	Hierarchical Clustering	2	0.405	31.138	0.713	Not Selected

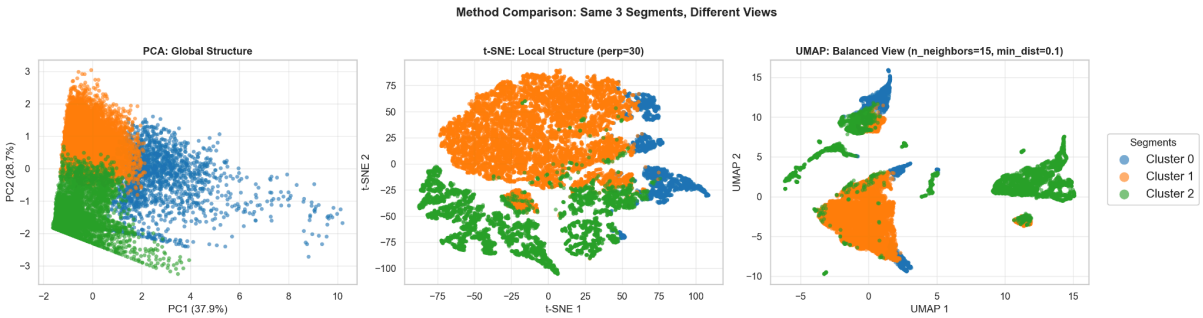
Annex 29

	Algorithm method	Number of Clusters (k)	Silhouette Score (↑ better)	Calinski-Harabasz Index (↑ better)	Davies-Bouldin Index (↓ better)	Preferred
0	K-means	2	0.421	6612.737	1.226	2nd selected
1	Hierarchical Clustering	4	0.250	4824.310	1.270	3rd selected
2	SOM + K-means	3	0.441	5892.855	1.096	1st selected

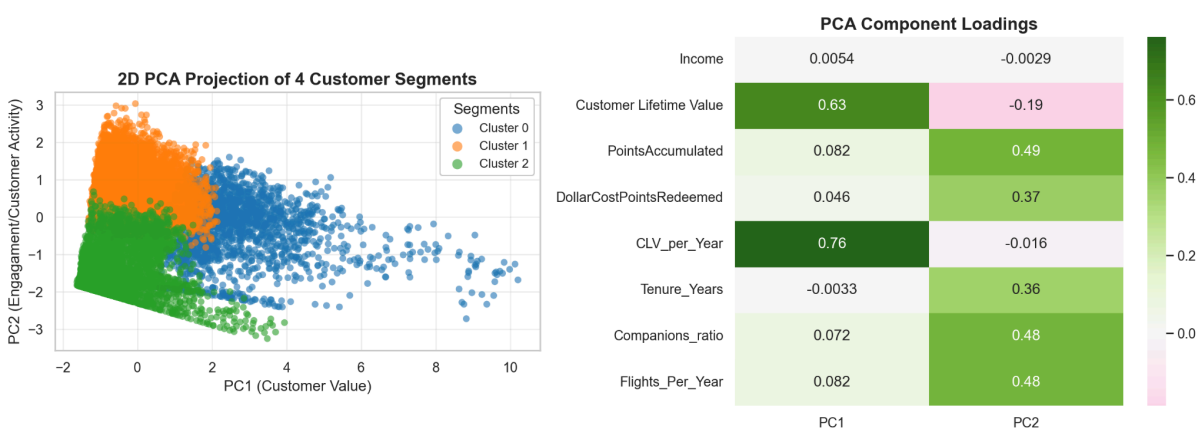
Annex 30

	Algorithm method	Number of Clusters (k)	Silhouette Score (↑ better)	Calinski-Harabasz Index (↑ better)	Davies-Bouldin Index (↓ better)	Preferred
0	K-means	2	0.484	18948.496	0.790	1st selected
1	Hierarchical Clustering	2	0.479	17891.382	0.776	2nd selected
2	SOM + Kmeans	2	0.469	18305.005	0.822	3rd selected

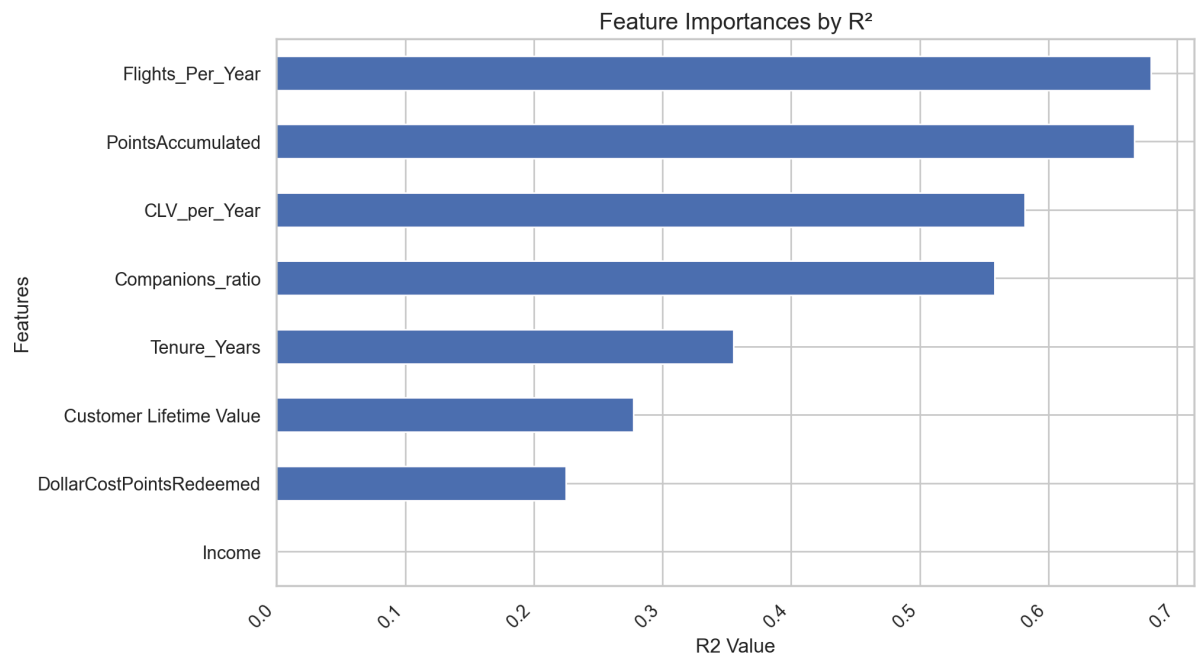
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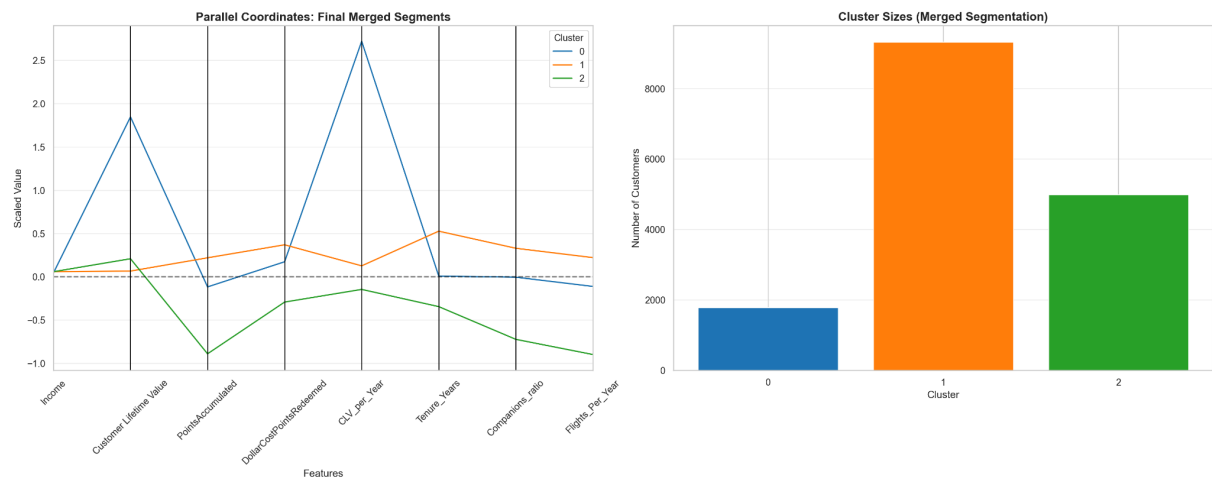
Annex 32



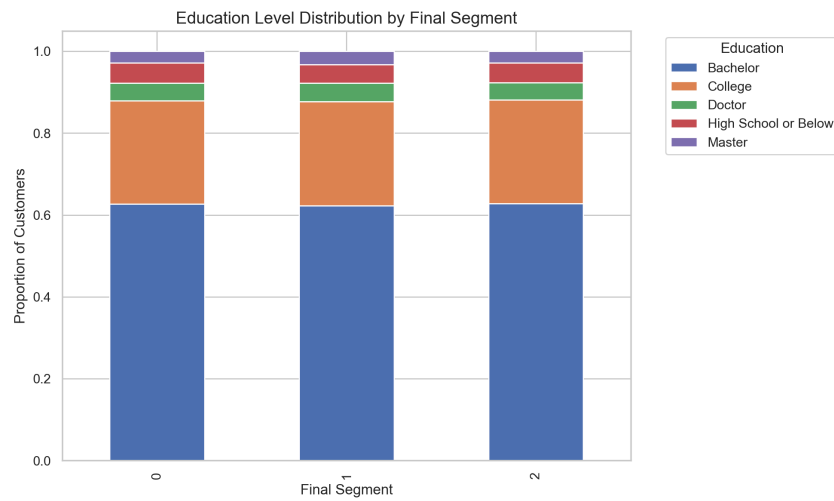
Annex 33



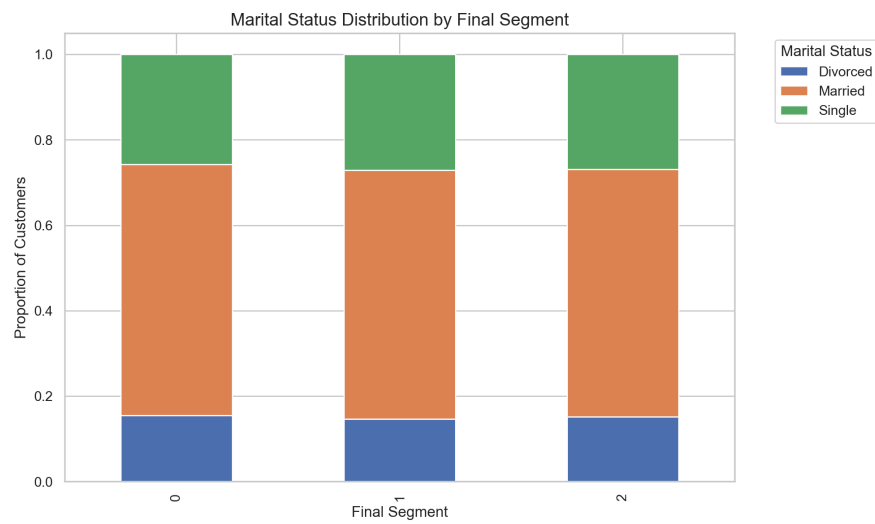
Annex 34



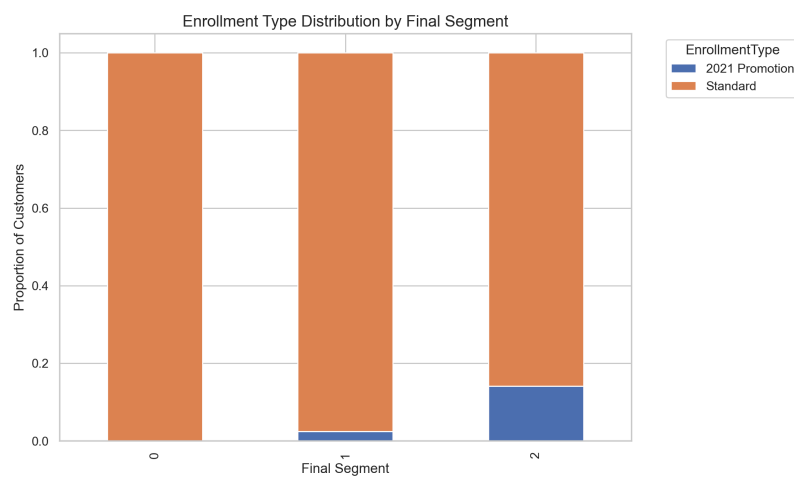
Annex 35



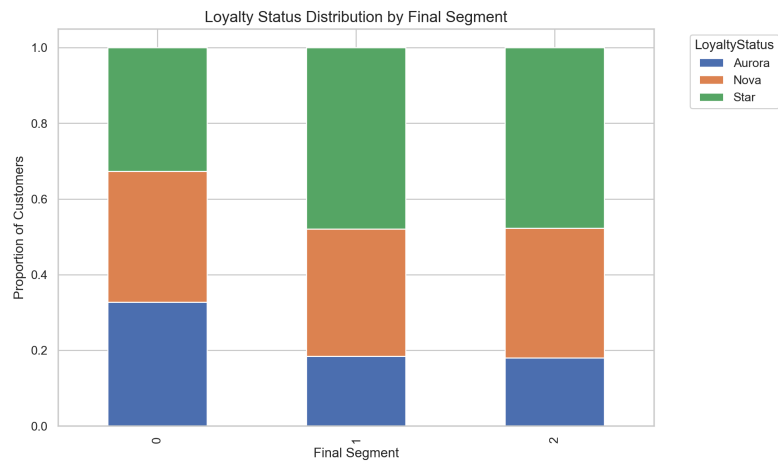
Annex 36



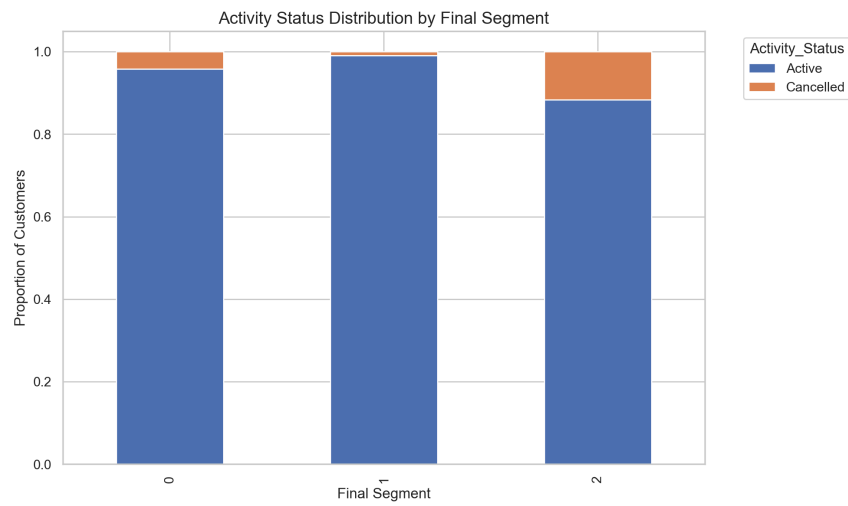
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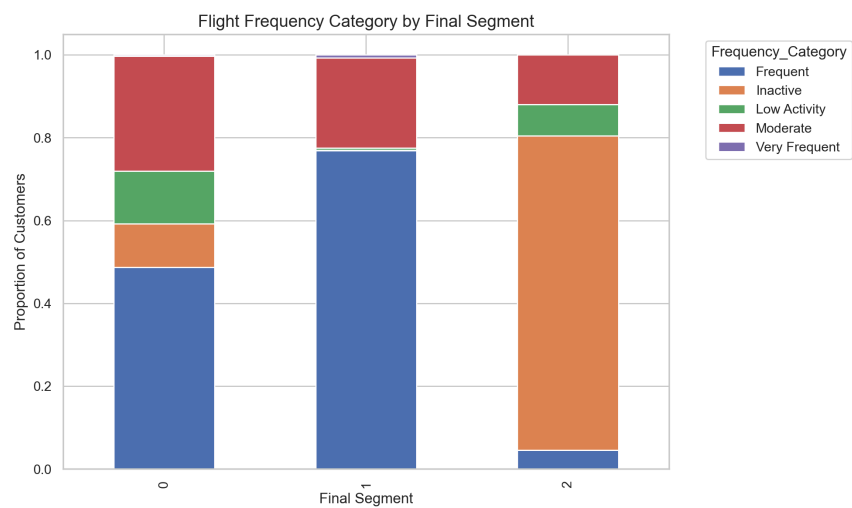
Annex 38



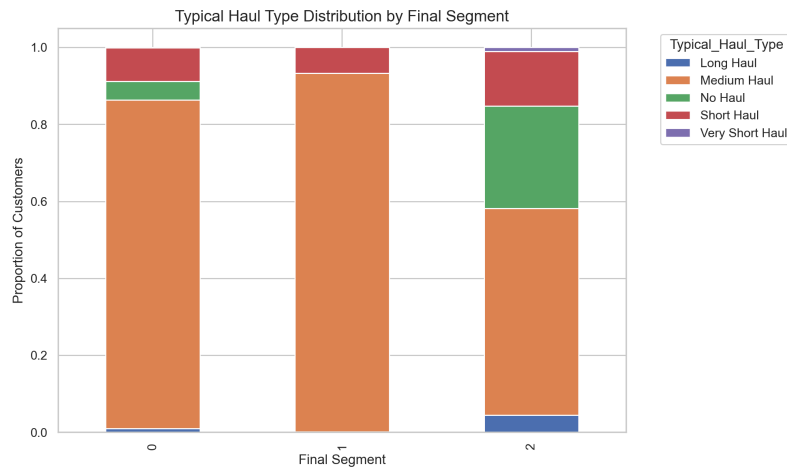
Annex 39



Annex 40



Annex 41



9. ANNEXES - STATEMENTS

AI Usage Statement

This report was prepared with the assistance of AI tools for specific purposes.

- **Tools Used:** ChatGPT (GPT-5), Gemini (2.5 Pro)
- **Purpose of Use:** Coding assistance to support debugging and error resolution, clarification of methodological and statistical concepts, and enhancement of wording.
- **Extent of Use:** AI was used to refine language, suggest formatting improvements, and enhance some strategic recommendations; all analysis and conclusions were completed manually by the team.
- **Verification:** All AI-generated content was **critically reviewed, edited, and verified** by the project team to ensure accuracy and originality.

Contribution Statement

Student Number	Name	Contributions
20221832	Andreia Fung	25%
20221889	João Bernardo	25%
20221950	Inês Caetano	25%
20222046	Rita Almeida	25%

All members participated in every part of the development of the report, code, and bonus components. We also all participated in team meetings, discussions of findings, decision-making and final report review.

Responsibility Statement

We certify that:

- This report represents our group's original work.
- All external sources and AI tools have been properly acknowledged.
- Each member has made a substantial contribution to the completion of this work.
- We collectively accept full responsibility for the content and integrity of this report.